avana

Walk without Pain
Live without Fear

Gentle, physiotherapist - designed exercises to relieve stiffness, strengthen your knees, and bring ease back to every step



Your 5-Minute Daily Routine to Move Freely Again...

Osteoarthritis:

It is commonly known as wear and tear disease, it's a condition in which the natural cushioning between the joints, i.e., cartilage, is damaged and wears away over time.

The bones of the joints rub more against one another with less shock absorbing benefits of cartilage. This results in pain, swelling, stiffness and decreased ability to move.

The treatment goal for OA of the knee includes relief of pain, inflammation, reduction of stiffness, improve the range of motion, mobility, and function.

The below exercises are to maximize the range of motion, flexibility, increase the strength and control of your knees.

These exercises will help you to return to daily activities and enjoy a more active, healthy lifestyle.

Talk to your Doctor or Physical Therapist about which exercises will best for you.

Strength:
Strengthening the muscles that support your knee will reduce stress on your knee joint. Strong muscles help your knee joint absorb shock.

Flexibility:
Stretching the muscles that you strengthen is important for restoring range of motion and preventing injury. Gently stretching after strengthening exercises can help reduce muscle soreness and keep your muscles long and flexible.

Target Muscles:
The muscle groups targeted in this conditioning program include:

- Quadriceps (front of the thigh)
- Hamstrings (back of the thigh)
- Abductors (outer thigh)
- Adductors (inner thigh)
- Gluteus Medius and gluteus maximus (buttocks)

Getting Started

Warmup: Before doing the following exercises, warm up with 5 to 10 minutes of low impact activity, like walking or riding a stationary bicycle.

Do not ignore pain: You should not feel pain during exercise. Talk to your doctor or physical therapist if you have any pain while exercising.

Ask questions: If you are not sure how to do an exercise, or how often to do it, contact your doctor or physical therapist.

Stretching Exercises

1. Heel Cord Stretch:

Repetitions
2 sets of 4

Days per week
6 to 7

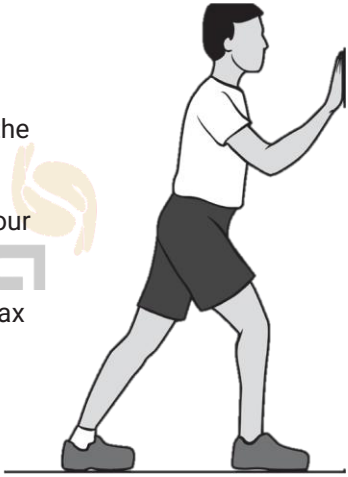
Main muscles worked:

Gastrocnemius-soleus complex. You should feel this stretch in your calf and into your heel.

Equipment needed: None

Step-by-step directions:

- Stand facing a wall with your unaffected leg forward with a slight bend at the knee. Your affected leg is straight and behind you, with the heel flat and the toes pointed in slightly.
- Keep both heels flat on the floor and press your hips forward toward the wall.
- Hold this stretch for 30 seconds and then relax for 30 seconds. Repeat.



Tip	Do not arch your back.

2. Standing Quadriceps Stretch:

Main muscles worked: Quadriceps. Gastrocnemius-soleus complex. you should feel this stretch in the front of your thigh.

Repetitions
2 to 3

Days per week
4 to 5

Equipment needed: None

Step-by-step directions:

- Hold on to the back of a chair or a wall for balance.
- Bend your knee and bring your heel up toward your buttock.
- Grasp your ankle with your hand and gently pull your heel closer to your body.
- Hold this position for 30 to 60 seconds.
- Repeat with the opposite leg.



Tip	Do not arch or twist your back.

3. Supine Hamstring Stretch _:

Main muscles worked: Hamstrings.

You should feel this stretch at the back of your thigh and behind your knee.

Equipment needed: None

Step-by-step directions:

- Lie on the floor with both legs bent.
- Lift one leg from the floor and bring the knee toward your chest. Clasp your hands behind your thigh below your knee.
- Straighten your leg and then pull it gently toward your head, until you feel a stretch. (If you have difficulty clasping your hands
- behind your leg, loop a towel around your thigh. Grasp the ends of the towel and pull your leg toward you.)
- Hold this position for 30 to 60 seconds.
- Repeat with the opposite leg.

Repetitions
2 to 3

Days per week
4 to 5



Tip	Do not put your hands at your knee joint and pull

Strengthening Exercises

4. Half Squads:

Repetitions
3 Sets of 10

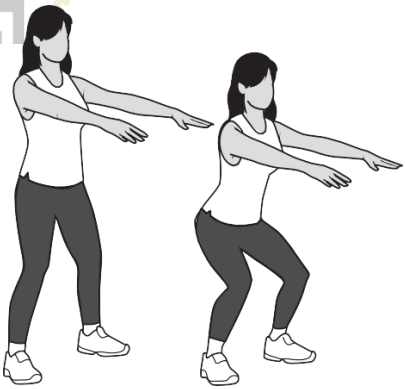
Main muscles worked: Quadriceps, gluteus, hamstrings.

You should feel this exercise at the front and back of your thighs, and your buttocks.

Days per week
4 to 5

Equipment needed: As the exercise becomes easier to perform, gradually increase the resistance by holding hand weights. Begin with 5-pound weights and gradually progress to a greater level of resistance, up to 10-pound weights.

Step-by-step directions:



- Lie on the floor with both legs bent.
- Stand with your feet shoulder distance apart. Your hands can rest on the front of your thighs or reach in front of you. If needed, hold on to the back of a chair or wall for balance.
- Keep your chest lifted and slowly lower your hips about 10 inches, as if you are sitting down into a chair.
- Plant your weight in your heels and hold the squat for 5 seconds.
- Push through your heels and bring your body back up to standing.

Tip	Do not bend forward at your waist.

5. Hamstring Curls:

Main muscles worked: Hamstrings.

You should feel this exercise at the back of your thigh

Repetitions
3 Sets of 10

Days per week
4 to 5

Equipment needed: As the exercise becomes easier to perform, gradually increase the resistance by adding an ankle weight. Begin with a 5-pound weight and gradually progress to a greater level of resistance, up to a 10-pound weight. If you have access to a fitness center, this exercise can also be performed on a weight machine. A fitness assistant at your gym can instruct you on how to use the machines safely.



Step-by-step directions:

- Hold onto the back of a chair or a wall for balance.
- Bend your affected knee and raise your heel toward the ceiling as far as possible without pain.
- Hold this position for 5 seconds and then relax. Repeat.
- Repeat with the opposite leg.

Tip	Flex your foot and keep your knees close together.

6. Calf Raises:

Main muscles worked: Gastrocnemius-soleus complex You should feel this exercise in your calf.

Equipment needed: Chair for support.

Repetitions
2 Sets of 10

Days per week
6 to 7

Step-by-step directions:

- Stand with your weight evenly distributed over both feet. Hold onto the back of a chair or a wall for balance.
- Lift your unaffected foot from the floor so that all your weight is placed on your affected foot.
- Raise the heel of your affected foot as high as you can, then lower.
- Repeat 10 times.



Tip	Keep your weight centered on the ball of your working foot.

7. Leg Extensions _:

Main muscles worked: Quadriceps.

You should feel this exercise at the front of your thigh.

Repetitions
3 Sets of 10

Days per week
4 to 5

Equipment needed: As the exercise becomes easier to perform, gradually increase the resistance by adding an ankle weight. Begin with a 5-pound weight and gradually progress to a greater level of resistance, up to a 10-pound weight. If you have access to a fitness center, this exercise can also be performed on a weight machine. A fitness assistant at your gym can instruct you on how to use the machines safely.



Step-by-step directions:

- Sit up straight on a chair or bench.
- Tighten your thigh muscles and slowly straighten and raise your affected leg as high as possible.
- Squeeze your thigh muscles and hold this position for 5 seconds. Relax and bring your foot to the floor. Repeat.

Tip	Do not swing your leg or use forceful momentum to lift it higher.

8. Straight-Leg Raises:

Main muscles worked: Quadriceps.

You should feel this exercise at the front of your thigh.

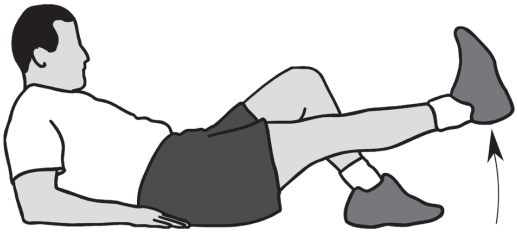
Repetitions
3 Sets of 10

Days per week
4 to 5

Equipment needed: As the exercise becomes easier to perform, gradually increase the resistance by adding an ankle weight. Begin with a 5-pound weight and gradually progress to a greater level of resistance, up to a 10-pound weight. If you have access to a fitness center, this exercise can also be performed on a weight machine. A fitness assistant at your gym can instruct you on how to use the machines safely.

Step-by-step directions:

- Lie on the floor with your elbows directly under your shoulders to support your upper body.
- Keep your affected leg straight and bend your other leg so that your foot is flat on the floor.
- Tighten the thigh muscle of your affected leg and slowly raise it 6 to 10 inches off the floor.
- Hold this position for 5 seconds and then relax and bring your leg to the floor. Repeat.



Tip	Do not tense up in your neck and shoulders.

9. Straight-Leg Raises (Prone) :

Main muscles worked: Hamstrings, gluteus.

You should feel this exercise at the back of your thigh and into your buttock.

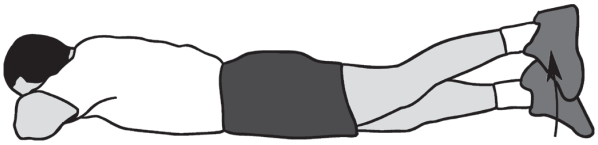
Repetitions
3 Sets of 10

Days per week
4 to 5

Equipment needed: As the exercise becomes easier to perform, gradually increase the resistance by adding an ankle weight. Begin with a 5-pound weight and gradually progress to a greater level of resistance, up to a 10-pound weight. If you have access to a fitness center, this exercise can also be performed on a weight machine. A fitness assistant at your gym can instruct you on how to use the machines safely.

Step-by-step directions:

- Lie on the floor on your stomach with your legs straight. Rest your head on your arms.
- Tighten your gluteus and hamstring muscles of the affected leg and raise the leg toward the ceiling as high as you can.



- Hold this position for 5 seconds.
- Lower your leg and rest it for 2 seconds. Repeat.

Tip	Keep your pelvic bones on the floor.

10. Hip Abduction:

Main muscles worked: Abductors, Gluteus

You should feel this exercise at your outer thigh and buttock.

Repetitions

3 Sets of 20

Days per week

4 to 5

Equipment needed: As the exercise becomes easier to perform, gradually increase the resistance by adding an ankle weight. Begin with a 5-pound weight and gradually progress to a greater level of resistance, up to a 10-pound weight.

Step-by-step directions:



- Lie on your side with your injured leg on top and the bottom leg bent to provide support.
- Straighten your top leg and slowly raise it to 45°, keeping your knee straight, but not locked.
- Hold this position for 5 seconds.
- Slowly lower your leg and relax it for 2 seconds. Repeat.

Tip Do not rotate your leg to raise it higher.

11. Hip Adduction:

Main muscles worked: Adductors.

You should feel this exercise at your inner thigh.

Equipment needed: As the exercise becomes easier to perform, gradually increase the resistance by adding an ankle weight. Begin with a 5-pound weight and gradually progress to a greater level of resistance, up to a 10-pound weight.

Repetitions
3 Sets of 20

Days per week
4 to 5

Step-by-step directions:

- Lie down on the floor on the side of your injured leg with both legs straight.
- Cross the uninjured leg in front of the injured leg.
- Raise the injured leg 6 to 8 inches off the floor.
- Hold this position for 5 seconds.
- Lower your leg and rest for 2 seconds. Repeat.



Tip	Place your hand on the floor in front of your stomach for support.

12. Leg Presses :

Main muscles worked: Quadriceps, hamstrings.

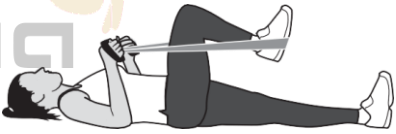
You should feel this exercise at the front of your hip, and the front and back of your thigh.

Repetitions
3 Sets of 10

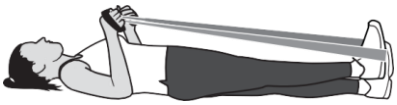
Days per week
4 to 5

Equipment needed: This exercise is best performed using an elastic stretch band of comfortable resistance. As the exercise becomes easier to perform, gradually increases the level of resistance. Do not use ankle weights with this exercise. If you have access to a fitness center, this exercise can also be performed on a weight machine. A fitness assistant at your gym can instruct you on how to use the machines safely.

Step-by-step directions:



- Place the center of the elastic band at the arch of your foot and hold the ends in each hand. Lie on the floor with your elbows bent.
- Tighten the thigh muscle of your affected leg and bring your knee toward your chest.
- Flex your foot and slowly straighten your leg directly in front of you, pushing against the elastic band.
- Hold this position for 2 seconds. Relax and bring your leg to the floor. Repeat.



Source:

AAOS American Academy of Orthopaedic Surgeons OrthoInfo

Tip	Keep your abdominals tight throughout the exercise.

Do's & Don'ts for people who are affected by.

Knee Osteoarthritis



Do's



Stay Active: Engage in low-impact exercises and activities like walking, swimming, or cycling to keep your knee joint mobile and strengthen the surrounding muscles.



Maintain a Healthy Weight: Excess weight can cause additional stress on your knee joints, so maintaining a healthy weight can help reduce. pain and improve mobility.



Use Assistive Devices: Consider using a cane or a walker if needed to reduce pressure on your knees when walking.



Exercise Regularly: Besides low-impact exercises, include stretching. and strengthening exercises for your quadriceps and hamstrings to support the knee joint.



Apply Heat or Cold Packs: Applying heat or cold packs to the affected Knee can help alleviate pain and reduce inflammation. Heat is typically more beneficial before exercise, and cold packs are more effective after activity.



Wear Comfortable and Supportive Shoes: Choose footwear with good arch support and cushioning to reduce impact on your knees while walking or standing.



Use Knee Braces or Sleeves: Some people find knee braces or compression sleeves helpful in providing additional support and stability to the joint.



Practice Good Posture: Maintain proper posture while sitting, standing, or walking to reduce strain on your knee joint.



Follow Your Doctor's Advice: Regularly visit your healthcare provider, follow their recommendations, and take prescribed medications as directed.

Don't



Avoid High-Impact Activities: Activities that put a lot of stress on the knees, like running or jumping, should be avoided, or minimized to prevent further damage.



Limit Prolonged Sitting or Standing: Try to avoid staying in the same position for extended periods, as it can lead to stiffness and discomfort.



Don't Overexert Yourself: While it's essential to stay active, avoid pushing yourself too hard during exercises or physical activities.



Avoid High Heels: High-heeled shoes can place excessive pressure on the knees. opt for comfortable, low-heeled footwear instead.



Don't Ignore Pain: If you experience persistent or severe pain, don't ignore it. Consult your doctor to assess the situation and adjust your treatment plan if necessary.



Avoid Excessive Lifting: Heavy lifting can strain your knees, so be cautious when lifting objects and ask for assistance when needed.



Don't Ignore Weight Gain: Gaining weight can worsen knee arthritis symptoms, so be mindful of your diet and exercise to maintain a healthy weight.

Remember that knee arthritis management should be personalized to your specific condition. Always consult your doctor or a healthcare professional for personalized advice and treatment recommendations based on your individual needs and medical history.

A Medical Education Initiative by

Avana Surgical Systemts Pvt.Ltd.,

No.91, Sundar Nagar 4th Avenue, Nandambakkam,
Chennai, Tamilnadu, India – 600 032.
Mail Id: info@avanasurgical.com
Mobile: +91 91500 64620
Phone: 044-2233 1061 /62 /63
Website: www.avanasurgical.com

Regional Offices:

Mumbai
The Summit Business Bay (Omkar) Office No. 606,
6th Floor, Andheri Kurla Road, Chakala,
Andheri East, Mumbai – 400 093.
Phone: 022-4970 0628

Bengaluru
No.52, 3rd Floor, Agastya Arcade,
80 feet Road, New BEL Rd, Devasandra Layout,
Bengaluru – 560 054, Karnataka, India
Phone: 080-2351 2259

New Delhi
Avana Medical Devices Pvt Ltd B6,
Qutab institutional Area,
New Delhi, Delhi – 110016
Phone: 011-4153 8222